

Greater Manchester Adult Hypertension Medication Pathway

Diagnosis & Assessment : ABPM or Average Home BP (Twice Am and Pm x 4 days)

Titrate every 4 Weeks after BP reassessment

Consider referral to community pharmacy for New Medication Service OR ABPM where available (tiny.one/commpharmacy)

TARGETS

Age <80; Home/Average < 135/85 mmHg or clinic < 140/90mmHg
Age 80+; Home/Average < 145/85 mmHg or clinic < 150/90 mmHg

Clinic BP ≥ 140/90 mmHg (age <80)
or ≥ 150/90 mmHg (age 80+)

Age < 40 - consider referral and secondary causes

Previous history of raised BP readings including in pregnancy, CKD, TIA, Stroke, Hypertensive retinopathy, sleep apnoea, proteinuria, diabetes, unexplained LVH

Take into account the person's motivation and wider situation to tailor support for self-management and lifestyle changes.

Advise reduction in salt & salty foods, reduce alcohol to < 15 units /week, weight to BMI < 25, DASH diet (tiny.one/dashdiet), exercise for 30+minutes x 3-5/week.

Baseline investigations: Lipids • U&Es • LFTs • TFT • HbA1c • ECG • Urine Dip • ACR

Age < 55 or T2 Diabetes

A

LISINOPRIL 10 mg OM

Not at target

Clinic BP ≥ 160/110 mmHg
OR
Home Avg BP ≥ 155/105 mmHg

A+C

Age 55+ or Black African or African Caribbean origin

C

AMLODIPINE 5 mg OM

Not at target

Combine LISINOPRIL 10mg OM with AMLODIPINE 5mg OM

(combination therapy is approximately 5 x more effective than doubling monotherapy)

Not at target

Consider pre-payment certificates for patients that pay for prescriptions

Not at target

Increase LISINOPRIL to 20mg OM
(See yellow box for monitoring requirements and eGFR change thresholds)

Not at target

Increase AMLODIPINE to 10mg OM

Not at target

A+c+D

ADD INDAPAMIDE 2.5mg OM

A+C+D

Not at target

A+C+D

INCREASE AMLODIPINE TO 10 mg, LISINOPRIL TO 20mg WITH INDAPAMIDE 2.5mg OM

Not at target

RESISTANT TO 3 DRUGS

>90% of patients resistant to 3 drugs are not taking them
CHECK CONCORDANCE

Reconsider salt, alcohol, other drugs incl. NSAIDS, Steroids, Additives eg Liquorice, Cancer Therapies;
? Coarctation, Sleep Apnoea, Obesity

FURTHER OPTIONS

K⁺ > 4.6 mmol/L : Doxazosin IR 2-4 mg OD, or Bisoprolol 2.5mg OD
K⁺ < 4.6 mmol/L : Spironolactone 12.5mg OM
or refer for specialist advice

Recheck 6 Months to 1 Year
Once stable

¹ Medication may require dose adjustment based on co-morbidities (e.g. renal/hepatic impairment) always check the BNF if uncertain.
Patients currently stable on their anti-hypertensives should not be switched due to this guidance.
This pathway is not applicable in pregnancy.

CKD

eGFR 20-45mL/min then consider maximum dose ACE/ARB and SGLT_i according to license.

eGFR 45-90mL/min + (Type 2 Diabetic or urine ACR >22.5) then consider maximum dose ACE/ARB and SGLT_i according to license

See GM CKD guide and [NICE CKS](https://www.nice.org.uk/guidance/NG208) for more information.

[NICE TA775](https://www.nice.org.uk/guidance/TA775) (Dapagliflozin for CKD) and [NICE TA942](https://www.nice.org.uk/guidance/TA942) (Empagliflozin for CKD)

Post Stroke/CKD and Urine ACR > 70mg/mmol: Aim for clinic BP <130/80 mmHg

A

ACEi/ARB

1st Line: LISINOPRIL¹ 10mg OM
(PERINDOPRIL ERBUMINE¹ 4mg OM)

Ramipril is shorter acting

2nd Line: CANDESARTAN¹ 8mg OM

IF ACE-I induced cough (more effective and safer than Losartan)

Use ARB if Black African or African Caribbean Origin.

Avoid in women of childbearing potential unless on effective contraception (use Amlodipine if trying to conceive).

Check U+E's before and 1-2 weeks after initiation & dose change. If eGFR decreases by < 25%, recheck levels after 1-2 weeks.

If eGFR decreases by > 25% or creatinine > 30% : Investigate for secondary causes and if persists despite these stop the ACE-i OR reduce dose to previously tolerated (recheck 5-7 days)

[Angiotensin-converting enzyme inhibitors | Prescribing information | Hypertension | CKS | NICE](https://www.nice.org.uk/guidance/TA775)

C

Calcium Channel Blocker

1st Line: AMLODIPINE 5mg OM
Amlodipine 5mg gives 80% of the effect of Amlodipine at 10mg

2nd Line: LERCANIDIPINE 10mg OM
If troublesome ankle swelling

D

Diuretic

1st Line: INDAPAMIDE¹ 2.5mg OM

Check U&E's before starting, and at regular intervals, within 2 months and ensure Na⁺ remains > 130 mmol/L, otherwise STOP

Recheck and, if improved, consider Bendroflumethiazide¹ 2.5mg OD
[Thiazide-like diuretics | Prescribing information | Hypertension | CKS | NICE](https://www.nice.org.uk/guidance/TA775)